```
========================================
INSERTING KEY: 20
========================================
Tree was empty. Node 20 set as Root.
   [Verification - Root Creation]:
     * Node (20) Predecessor -> NULL
     * Node (20) Successor   -> NULL
     * Verified: Left link is a thread pointing to Predecessor (NULL)
     * Verified: Right link is a thread pointing to Successor (NULL)

========================================
INSERTING KEY: 10
========================================
Attached 10 as LEFT child of 20
Thread Updates:
 -> 10->left  thread now points to Predecessor: NULL
 -> 10->right thread now points to Successor:   20
 -> Parent 20->lthread set to FALSE (now structural link)
   [Verification - Newly Inserted Node Validation]:
     * Node (10) Predecessor -> NULL
     * Node (10) Successor   -> 20
     * Verified: Left link is a thread pointing to Predecessor (NULL)
     * Verified: Right link is a thread pointing to Successor (20)
   [Verification - Parent Node Link Validation]:
     * Node (20) Predecessor -> 10
     * Node (20) Successor   -> NULL
     * Verified: Right link is a thread pointing to Successor (NULL)

========================================
INSERTING KEY: 30
========================================
Attached 30 as RIGHT child of 20
Thread Updates:
 -> 30->left  thread now points to Predecessor: 20
 -> 30->right thread now points to Successor:   NULL
 -> Parent 20->rthread set to FALSE (now structural link)
   [Verification - Newly Inserted Node Validation]:
     * Node (30) Predecessor -> 20
     * Node (30) Successor   -> NULL
     * Verified: Left link is a thread pointing to Predecessor (20)
     * Verified: Right link is a thread pointing to Successor (NULL)
   [Verification - Parent Node Link Validation]:
     * Node (20) Predecessor -> 10
     * Node (20) Successor   -> 30

--- Base Tree Status ---
Current Inorder Sequence: 10 20 30
```

```
--- Base Tree Status ---
Current Inorder Sequence: 10 20 30

========================================
INSERTING KEY: 2
========================================
Attached 2 as LEFT child of 10
Thread Updates:
 -> 2->left  thread now points to Predecessor: NULL
 -> 2->right thread now points to Successor:   10
 -> Parent 10->lthread set to FALSE (now structural link)
   [Verification - Newly Inserted Node Validation]:
     * Node (2) Predecessor -> NULL
     * Node (2) Successor   -> 10
     * Verified: Left link is a thread pointing to Predecessor (NULL)
     * Verified: Right link is a thread pointing to Successor (10)
   [Verification - Parent Node Link Validation]:
     * Node (10) Predecessor -> 2
     * Node (10) Successor   -> 20
     * Verified: Right link is a thread pointing to Successor (20)

========================================
INSERTING KEY: 45
========================================
Attached 45 as RIGHT child of 30
Thread Updates:
 -> 45->left  thread now points to Predecessor: 30
 -> 45->right thread now points to Successor:   NULL
 -> Parent 30->rthread set to FALSE (now structural link)
   [Verification - Newly Inserted Node Validation]:
     * Node (45) Predecessor -> 30
     * Node (45) Successor   -> NULL
     * Verified: Left link is a thread pointing to Predecessor (30)
     * Verified: Right link is a thread pointing to Successor (NULL)
   [Verification - Parent Node Link Validation]:
     * Node (30) Predecessor -> 20
     * Node (30) Successor   -> 45
     * Verified: Left link is a thread pointing to Predecessor (20)

--- Final Tree Status ---
Current Inorder Sequence: 2 10 20 30 45

--------------------------------
Process exited after 0.09507 seconds with return value 0
Press any key to continue . . .
```

23°C
Mostly clear
Search
ENG
IN
10:50 PM
16-10-2026